

Answer ALL questions.

Write your answers in the spaces provided.

- 1** Coir fibres are collected from coconuts.

The photograph shows part of a coconut with some of these fibres.



(Source: © Mohammed Anwarul Kabir Choudhury / Alamy Stock Photo)

- (a) Name **two** types of fibre cells found in a plant.

(2)

1

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2

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(b) A student investigated the effect of length on the tensile strength of these fibres.

(i) State what is meant by the term **tensile strength**.

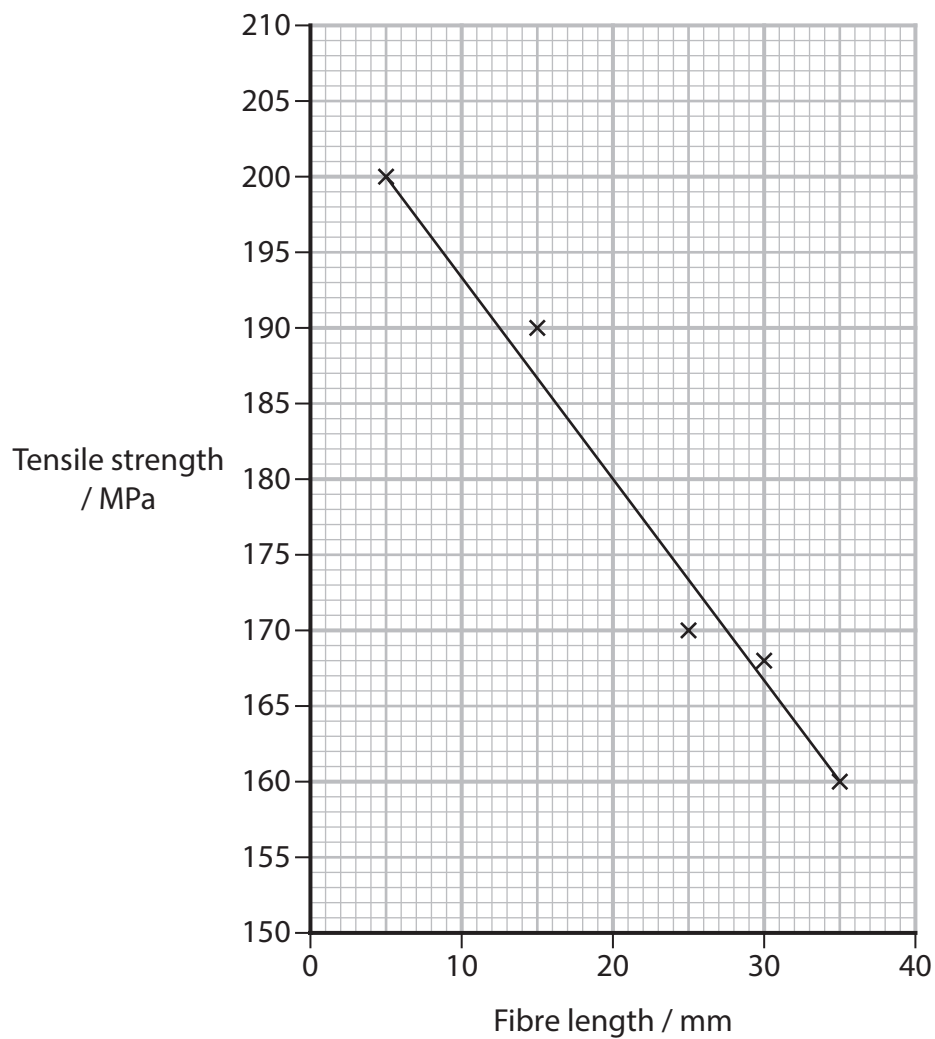
(1)

(ii) Describe how to investigate the relationship between the length of these fibres and their tensile strength.

(4)



(c) The graph shows the results of a similar investigation.



(i) Draw a table to show all the results shown in the graph.

(3)



(ii) Calculate the gradient of the line of best fit shown in this graph.

(1)

Answer

(iii) Write the equation for the line of best fit shown in this graph.
Use the graph and your answer to part (ii) to write this equation.

(2)

Equation for this line of best fit

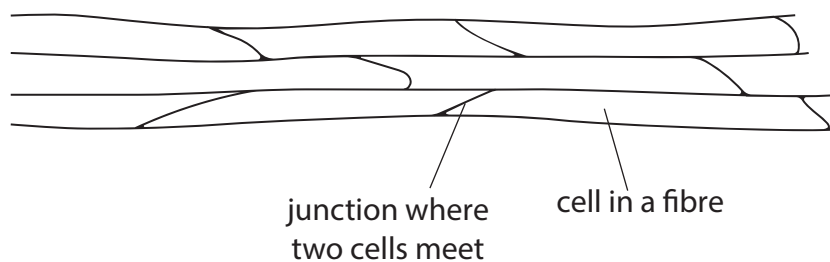
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(d) The drawing shows the structure of part of a plant fibre.



The table shows the gradients for similar graphs for three other types of fibre.

Fibre	Gradient
bamboo	-3.12
brown coir	-1.25
jute	-1.16

(i) State the relationship represented by the gradient.

(1)

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- (ii) Suggest how differences in the structure of these three fibres could affect their tensile strength.

Use the information in the graph, the drawing and the table to support your answer.

(3)

(Total for Question 1 = 17 marks)

